

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	02 November 2025
Team ID	NM2025TMID07694
Project Name	Educational Organisation Using ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

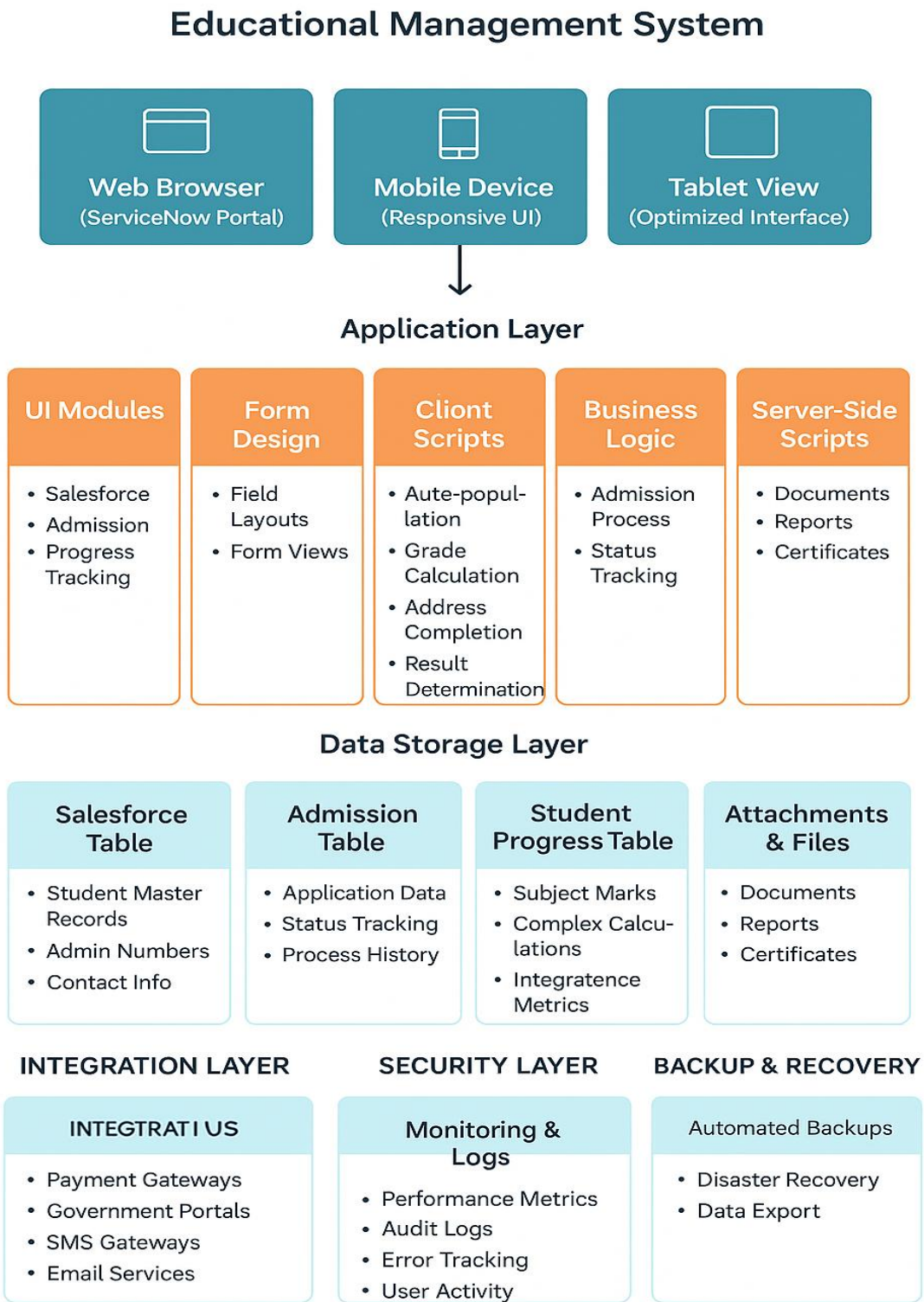


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	School staff interacts via ServiceNow web portal	ServiceNow Now Platform UI
2	Application Logic-1	Auto-populates student details from reference fields	Client Scripts (JavaScript)
3	Application Logic-2	Calculates totals, percentages and determines results	Client Scripts (JavaScript)
4	Application Logic-3	Auto-fills address fields based on pincode input	Client Scripts (JavaScript)
5	Application Logic-4	Manages admission process workflow and status tracking	ServiceNow Process Flow
6	Database	Stores student, admission, and progress data	ServiceNow Tables
7	Cloud Database	Managed by ServiceNow cloud infrastructure	ServiceNow Cloud Database
8	File Storage	Stores student documents and reports	ServiceNow Attachment Tables
9	External API-1	(Future) Integration with payment gateways for fee collection	REST API
10	External API-2	(Future) Integration with government educational portals	SOAP/REST API
11	Infrastructure	Fully hosted on ServiceNow SaaS platform	ServiceNow Cloud (SaaS)

Table-2: Application Characteristics

S.No	Component	Description	Technology
1	Open-Source Frameworks	Limited to ServiceNow ecosystem; uses JavaScript for customization	JavaScript, ServiceNow APIs
2	Security Implementations	Role-based access control, table-level security, form security	ServiceNow ACLs, Roles, User Criteria
3	Scalable Architecture	Cloud-native SaaS platform with automatic scaling	ServiceNow Multi-tenant Architecture
4	Availability	High availability through ServiceNow's enterprise cloud	ServiceNow Load-balanced Instances
5	Performance	Optimized through client-side scripts and efficient database queries	GlideRecord, Client Scripts, UI Policies
6	Data Integrity	Automated validation and calculation scripts ensure data accuracy	Business Rules, Client Scripts
7	User Experience	Responsive and intuitive interface for administrative staff	ServiceNow Form Design, UI16/UI17